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1- HIGH ORDER AUTOSHIM REFERENCE FIELD MAPS CREATION

1-1 Overview

The High Order Autoslim feature allows the user to shim over a region-of-interest (ROI) taking advantage of the High Order Shim (HOS) coils. This feature is likely to improve the local field homogeneity for use in applications such as fat suppression, spectroscopy, spiral scanning and functional imaging. The calculations performed in the High Order Autoslim procedure require knowledge of the change in current behaviour of each individual shim coil. Hence the contribution of each one is measured using a uniform phantom. This information is stored in the Reference Field Maps, which need to be calculated once per system installation. These Field Maps would need to be recalculated if the gradient coil is moved or replaced.

The basic process which runs automatically, is described in the instructions below, and comprises:

1. Running a set of scans identical to the scan used in the High Order Autoslim Feature on a uniform phantom, varying the currents on each of the coils, one at a time, by a prescribed amount.
2. Reconstructing the data and extracting the phase information that reflects the magnetic field variations.
3. Packing the data into a file (the reference field) that can be read by the High Order Autoslim feature.

1-2 Prerequisites

- Software version is 9.0 or higher.
- Verify patch Signa LX Patch_90_M4_03 has been loaded.
- System shim must be within specification.
- The system should be fully calibrated and ready for clinical scanning. Keep a record of **both** final sets of X, Y, Z gradient shim offset values from the LVShim Gradient Shim calibration.
- Communication with the Shim supply is working. The Shim power supply is connected with the supplied RS232 cable from the rear panel **COM 1** connector of the supply to the serial **port 2** on the Signa octane computer (SCSI Serial expansion unit). Verify functionality by opening a C-Shell, and typing **setzero**, followed by RETURN. The window should then scroll with five high order shim channels set to zero. If not, refer to the troubleshooting section to identify problem and correct it prior to continuing.
- Verify by opening a C-Shell and typing **cd /w/config** followed by RETURN, then typing **jot prescan_config.cfg** followed by RETURN, that the last line of this file has the field "**hos_exists=1**", not "0". If the value is "0" rather than "1", change the value to "1". Push "**File**", and leave cursor on "**Save**" if updated. Then close file by pushing on "**File**", followed by "**Exit**". If **hos_exists** was updated to 1, **reboot** Signa.
- Once only, prior to first run, open a C-Shell and type **cd /usr/g/bin** followed by RETURN, then type **create_howgood_lib** (with underlines included) followed by

RETURN. A response will be returned saying the directories already exist or they were created.

- Body SPT Sphere phantom (Model No. 2125244 identified as short loader in parts manual) mounted **without its cylindrical loader** for small FOV (<24cm) maps.
- LVShim phantom (Model No. 2135649) + its short nesting plate (Model No. 2125247-10) for large FOV (<48cm) maps.

1-3 Field map files creation procedure

There are four Reference Field maps to be generated for the TwinSpeed scanner. They are:

1. **FOV=24**, using the RF **BODY** coil and the **WHOLE-BODY** (WB) gradient coil
2. **FOV=24**, using the RF **BODY** coil and the **ZOOM** (ZM) gradient coil
3. **FOV=48**, using the RF **BODY** coil and the **WHOLE-BODY** (WB) gradient coil
4. **FOV=48**, using the RF **BODY** coil and the **ZOOM** (ZM) gradient coil

These four are designed so as to accommodate all possible anatomical regions, with smaller central regions, such as the head, neck, knees, using the FOV=24 maps, and any other region, either with large FOV or off-center, using the FOV=48 maps. Each one is duplicated for the different gradient coils.

The procedure for building the reference maps is the same for all four, with the exception of the phantom used. The full LVShim phantom is used for generating the large 48cm FOV reference maps. The Body SPT Sphere without the loader shell is used to generate the 24cm FOV reference maps. The Body SPT phantom is 27cm diameter that is larger than the 24cm FOV used but does not suffer from any detrimental wrap effects in the final reference maps. This generates the largest usable shim volume possible for use at 24cm fov scans. This is important since only anatomy inside the shim volume selected can be shimmed properly. Each map is generated separately, and does not have to be run in the same session.

1-3-1 Creating Calibration Files

1. Place the body SPT phantom in the center of the RF coil, and without the cylindrical loader. Use the following table for the correct setup, and for the recommended order to run the tests.

Table 1
High Order Shim File Calibration

GE/other series	Map	Gradient Coil	RF coil to use	Phantom to use
1	FOV=24	WHOLE-BODY	BODY	Body SPT Sphere
3	FOV=24	ZOOM	BODY	Body SPT Sphere
2	FOV=48	WHOLE-BODY	BODY	LVShim
4	FOV=48	ZOOM	BODY	LVShim

2. Run **TPS Reset** from the Service desktop if Signa was not re-booted in Section 1-2.
3. Landmark the center of the phantom, and Advance to Scan.
For the two 24 fov series make sure the SPT sphere is centered by selcting new patient with the Patient ID geservice, 111 lbs, Body coil, Head first, Supine, 3-plane 2D, Whole, Pulse seq: localizer 32 FOV, 10 slice thick, 0 skip, R/L A/P S/I = 0, slices 1, 256x256, 1 nex. Select "Save Series", then Scan. After the scan is complete select the series from the Image Browser. Turn on the **Grid** and check for phantom centering. The phantom should be centered within 10 mm in all three planes. Adjust pads and isocenter if necessary to center the phantom in the fov. Then **End Exam** and continue with the 24 fov reference map generation.
4. Open a C-Shell command window, from the Service Desktop.
5. Type **cd /usr/g/bin** followed by RETURN, then **makereference**, followed by RETURN.

Note

Monitor the information in the C-Shell during the *makereference* procedure for "communication timeout" errors. If errors are noted check cable between HOS power supply and host. If problems persist cycle power to the HOS power supply.

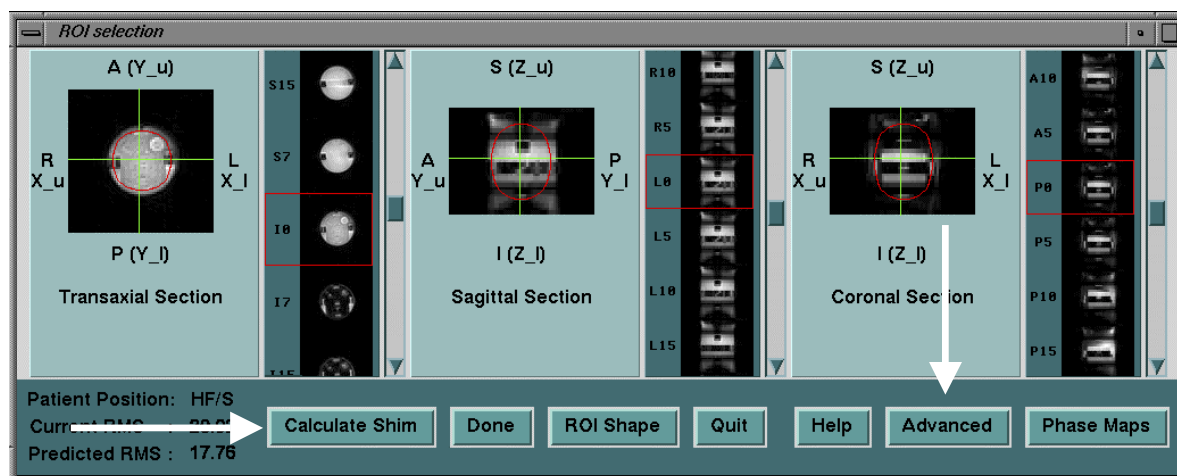
6. Select the FOV=24 followed by RETURN. FOV=48 will be done on subsequent iterations.
7. Select the gradient coil to be used, by typing **W** or **Z**, followed by RETURN.
8. Ensure that the phantom is in position, and then confirm with RETURN. The 24 FOV scans will complete in less than 4 minutes and the 48 FOV in less than 10 minutes each.

1-3-2 Creating Default Current Settings

The following procedure will create a default setting for the High Order Shim Supply. The default settings will only be used with the *Restore Defaults* button is pressed on the console.

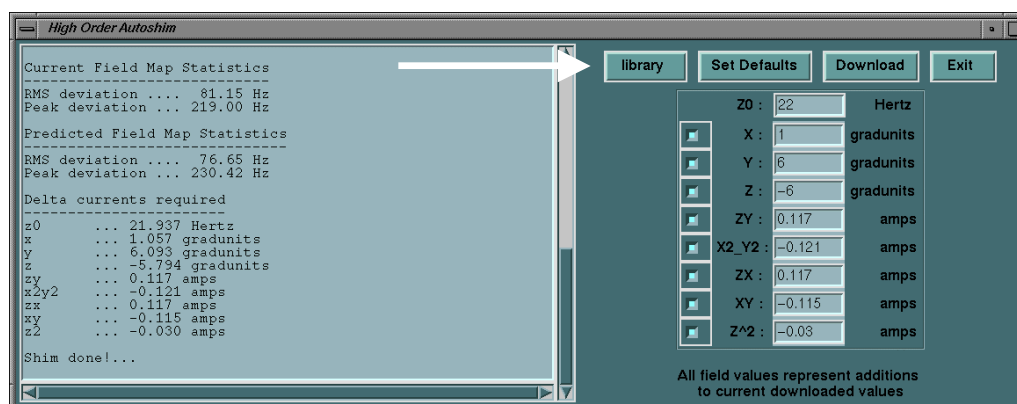
1. Once complete, enter **New Patient** from the Scan desktop, with patient ID as **mr**.
2. Select from **Service/other**, the protocol called **HighOrderAutoshimBody**. Select the series description that matches the reference map just generated.
3. **Save Series** this series without making any changes, and choose **ACCEPT**.
4. After download, press **SCAN**.

5. When the scan is finished, the High Order Shim (HOS) processing window will open. (takes approx 5-10 seconds after a 24cm fov series and approx 45 seconds for the 48cm fov series) See Illustration 1.

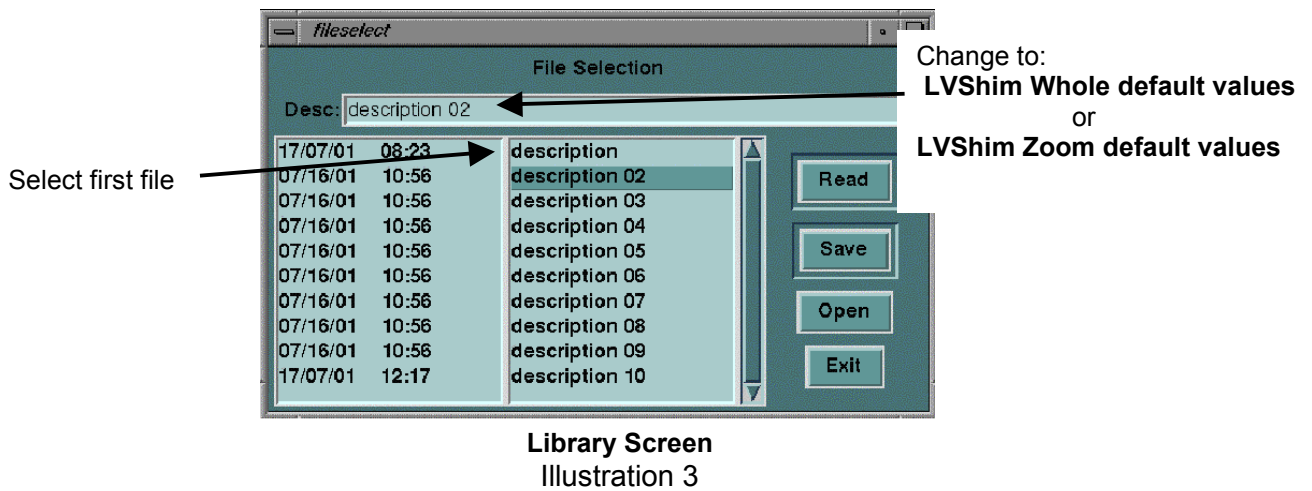


HOS Processing Window
Illustration 1

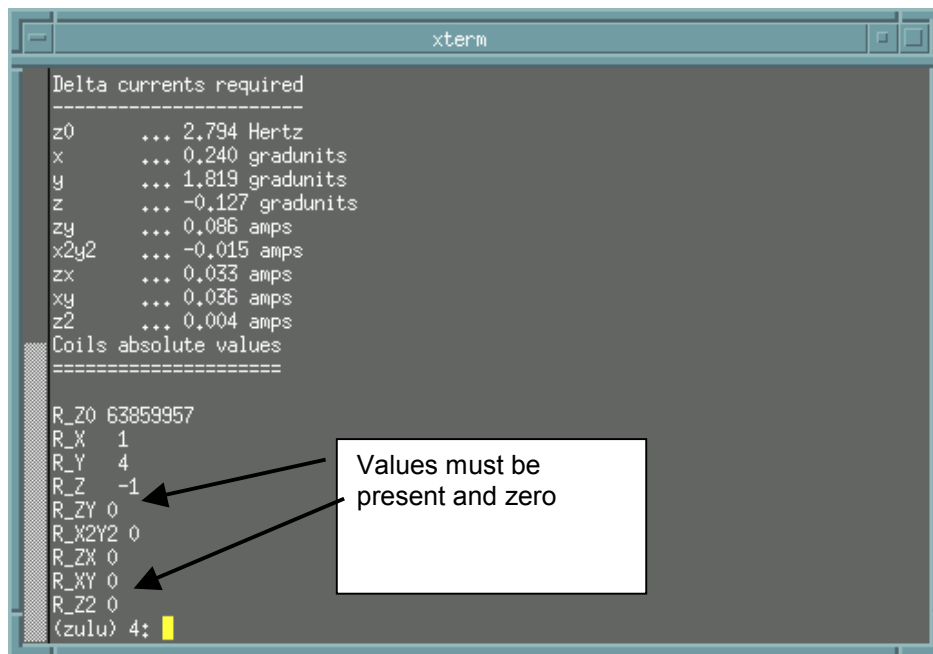
6. Center the ROI inside the phantom then click on **Calculate Shim**.
7. Click on **Advanced**, and then **Library**. See Illustrations 2 and 3 below.
8. Select the first file, then go to the **Desc** field, and change the name to **LVShim WHOLE (or ZOOM) default values**, depending on the coil type selected in Step 7. See Illustration 3.



Advanced Screen
Illustration 2



9. Click on **Save**.
10. When **Advanced** window given again, click a second time on **Library**, and select the first file, with name **LVShim WHOLE (or ZOOM) default values**.
11. Select **Open**, and update in the editor the "Coil Absolute Value" section **R_X**, **R_Y** and **R_Z** fields [scroll down to almost the end of the file] with the gradient shim values recorded from the LVShim calibration for the correct Gradient coil. Ensure that all five **R_high order shims** fields have a value of zero. **Note: scroll by using the left mouse button in the left scroll bar**. See Illustration 4.



R values
Illustration 4

Note

The values for the five High Order Shims may not be zero. If this is the case, exit the Library screen and re-select the Library option and recheck the values. They should be zero.

12. Click on **Save**, then **Quit** from the editor.
13. Click **Exit** to close the Library screen and then **Quit** from the basic screen, followed by **OK** to close the shim user interface.

Note

The 48CM Zoom mode may fail due to dark images. If this occurs try increasing the R1 value by one count.

14. Repeat steps 1 to 8 in Section 1-3 for the other coil type as given in Table 1. If Zoom was selected you must run Whole. Steps 9 to 21 need only be run once per Gradient coil, i.e. during the first run with **GradMode=WHOLE** and likewise for **GradMode=ZOOM**. Suggest doing this for the two 24cm fov series.
15. Change phantom from SPT sphere to the LVShim phantom. Repeat steps 1 to 8 for the 48cm FOV for both coil types of Zoom and Whole.
16. Perform **SAVE INFO** from the **Install GUI** to save the Reference Field maps onto an MOD.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	August 2, 2001	Jeremy Gerber	Revised from Signa 3T for High Order Shim option in TwinSpeed scanner with Leo1 release.
1	August 6, 2001	Jeremy Gerber	Updated with C-Shell functions
2	August 8, 2001	Jeremy Gerber	Updated with help screen
3	August 9, 2001	Jeremy Gerber	Corrected error in Library function
4	August 14, 2001	Jeremy Gerber	Corrections to Library definitions
5	August 16, 2001	Jeremy Gerber	Updated basic and Advanced screens
6	August 29, 2001	Jeremy Gerber	Updated units, A to mA, DAC to mA
7	Sep 2, 2001	Jeremy Gerber	Updated screens and reference field creation order
8	Sep 5, 2001	Jeremy Gerber	Updated sections 1-2 and 1-3.
9	Sep 5, 2001	Jeremy Gerber	Updated location of ho.log file.
10	Sep 13, 2001	Jeremy Gerber	Updated section 1-3.
11	Oct 23, 2001	Jeremy Gerber	Updated sections 1-2 (LVShim phantom), 2 (details), 3 (mA to DAC units)
12	Oct 24, 2001	Jeremy Gerber	Updated section 3 (Octane to Shim names)
13	Nov 15, 2001	Jeremy Gerber	Updated basic, Advanced screens, and Table 2-1.
14	Nov 26, 2001	Steve Merritt	Updated for changes from 32cm to 24cm and use of body coil and SPT ball.
15	Nov 30	Pete Kargard	Inserted Service documentation format.
16	Dec 5	Pete Kargard	Changed entire procedure into 2 sections for creating cal files and creating default settings.
17	Dec 10, 2001	Pete Kargard	Feedback from field site.